



Advanced Internet of Things Technology

Week 3: Edge Intelligence II – Collaborative Learning, Federated Learning, and Privacy in IoT

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Week 3 Overview



Positioning of Week 3 in the course

- Week 1 introduced Digital Twin concepts and system vision.
- Week 2 focused on TinyML and device-level edge intelligence.
- Week 3 moves from isolated intelligence to collaborative intelligence.

Why this lecture is necessary

- Single-device intelligence cannot scale to large IoT systems.
- Data, models, and decisions are distributed across layers.
- Coordination, learning, and trust become dominant challenges.

Objectives

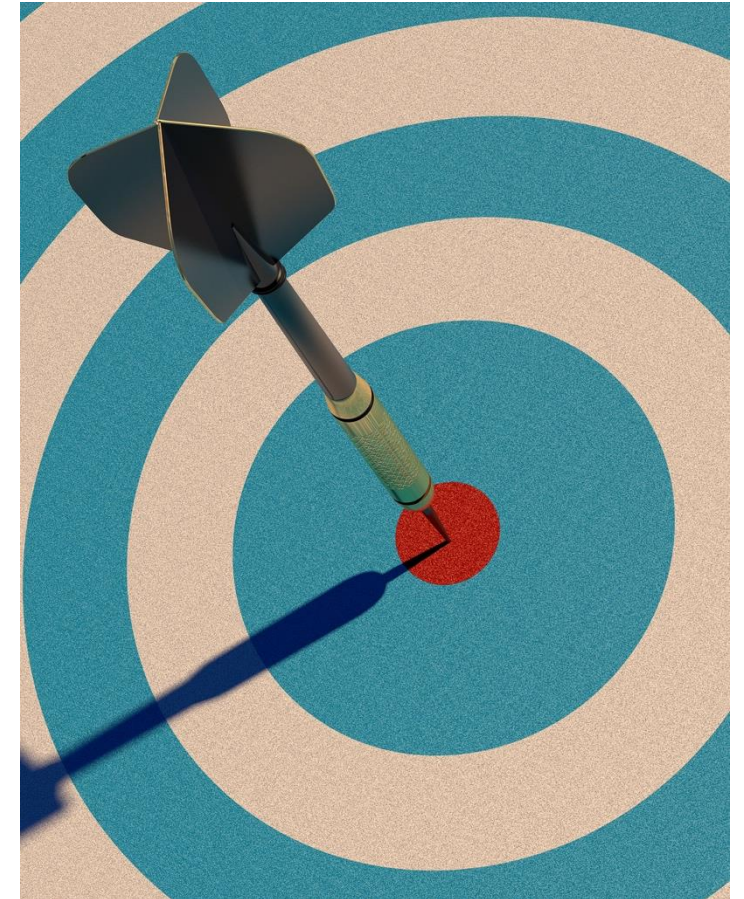


Core message of Week 3

- Intelligence must be collaborative rather than isolated.
- Learning must occur across cloud, edge, and terminals.
- Privacy and trust are architectural constraints.

Learning objectives

- Understand Cloud–Edge–Terminal collaborative intelligence.
- Understand Federated Learning as distributed optimisation.
- Reason about orchestration and resource management.
- Treat privacy and security as first-class design factors.



Part 1 – Cloud–Edge–Terminal Collaborative Intelligence Architecture

3-Layer Collaborative Intelligence Architecture



From cloud-centric to collaborative intelligence

- Traditional IoT relies heavily on cloud-centric intelligence.
- Cloud-only intelligence suffers from latency and bandwidth limits.
- Collaborative intelligence distributes roles across layers.

Overview of Cloud–Edge–Terminal architecture

- Terminal layer interacts with the physical environment.
- Edge layer provides low-latency intelligence and coordination.
- Cloud layer provides global learning and optimisation.

Intelligence placement as a design problem

- Where intelligence is placed affects latency and energy.
- Placement decisions affect privacy and data exposure.
- Optimal placement depends on application requirements.

Terminal layer – devices and sensors

Functional role of terminal layer

- Data acquisition from physical processes.
- Local preprocessing and inference.
- Immediate actuation and response.

Benefits of terminal-level intelligence

- Ultra-low latency reactions.
- Improved privacy by keeping raw data local.
- Reduced communication overhead.

Constraints at the terminal layer

- Limited memory and computing capability.
- Strict energy constraints.
- Limited model complexity.

Edge layer – gateways and edge servers



Functional role of the edge layer

- Aggregation of data and intelligence from terminals.
- Low-latency inference for groups of devices.
- Coordination among multiple terminals.

Edge intelligence capabilities

- More compute and memory than terminals.
- Support for local model adaptation.
- Short-term learning and caching.

Edge layer challenges

- Resource contention among services.
- Intermittent connectivity to cloud.
- Need for lightweight orchestration.

Cloud layer – global intelligence



Functional role of the cloud layer

- Large-scale model training.
- Global model aggregation and optimisation.
- Long-term analytics and system monitoring.

Cloud and digital twins

- Cloud maintains global digital twin states.
- Supports prediction and optimisation.
- Coordinates policies across the system.

Repositioning the cloud

- Cloud is not eliminated but redefined.
- Heavy computation and coordination stay in cloud.
- Time-critical decisions move closer to the edge.

Part 2 – Enabling Technologies for Collaborative Intelligence

Containerisation and lightweight virtualisation



Role of containerisation

- Package AI services consistently.
- Enable portability across cloud and edge.
- Simplify deployment and updates.

Network virtualisation and slicing

- Abstract physical network resources.
- Support multiple AIoT services concurrently.
- Provide isolation and QoS guarantees.

Edge-specific constraints

- Full cloud orchestration is too heavy.
- Edge requires lightweight runtimes.
- Offline autonomy must be supported.

Edge orchestration frameworks



Motivation for edge orchestration

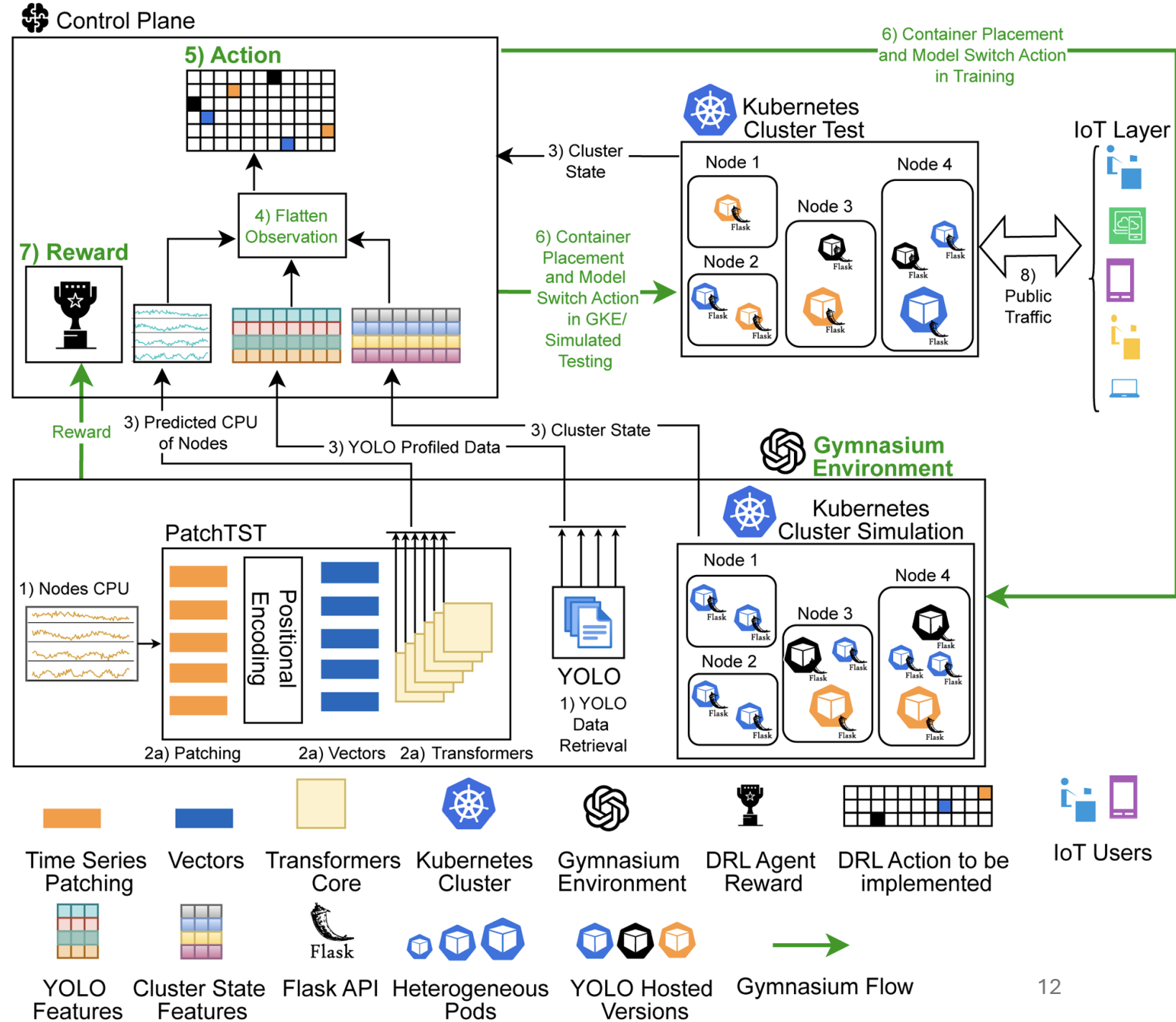
- Manage distributed AI lifecycle.
- Handle intermittent connectivity.
- Enable autonomous edge operation.

Core orchestration functions

- Service deployment and migration.
- Monitoring and recovery.
- Coordination with cloud controllers.

AI-Driven Joint Container Management

B. Ali, M. Golec, S. S. Gill, F. Cuadrado and S. Uhlig,
 "EdgeAIBus: AI-Driven Joint Container Management and Model Selection Framework for Heterogeneous Edge Computing," in *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 11, pp. 2412-2424, Nov. 2025, doi: 10.1109/TPDS.2025.3602521



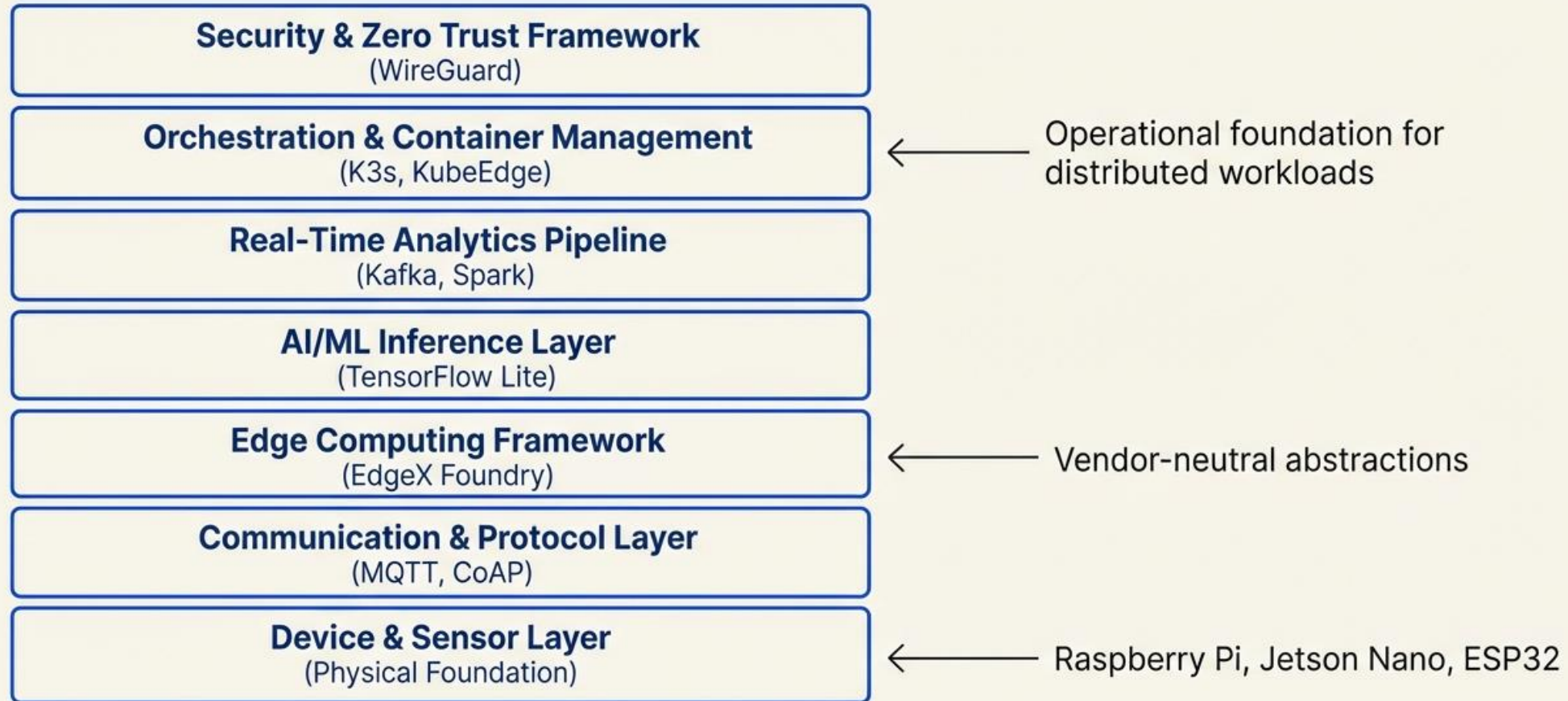
Role of SDN

- Centralised control of distributed networks.
- Dynamic routing and traffic prioritisation.
- Support latency-sensitive AI traffic.

Interaction with AI workloads

- Network state influences AI placement decisions.
- AI workload requirements influence routing.
- Joint optimisation of communication and computation.

A 7-Layer Blueprint for Production-Grade Edge Systems.



This stack balances computational efficiency with operational manageability.
NotebookLM

Part 3 – Federated Learning Fundamentals

Part 3 – Federated Learning Fundamentals



Why Federated Learning is needed in AIoT

- IoT data is naturally distributed across devices.
- Centralising data violates privacy and increases bandwidth cost.
- Regulations often prohibit raw data sharing.
- Devices experience heterogeneous environments and data distributions.

Federated Learning paradigm

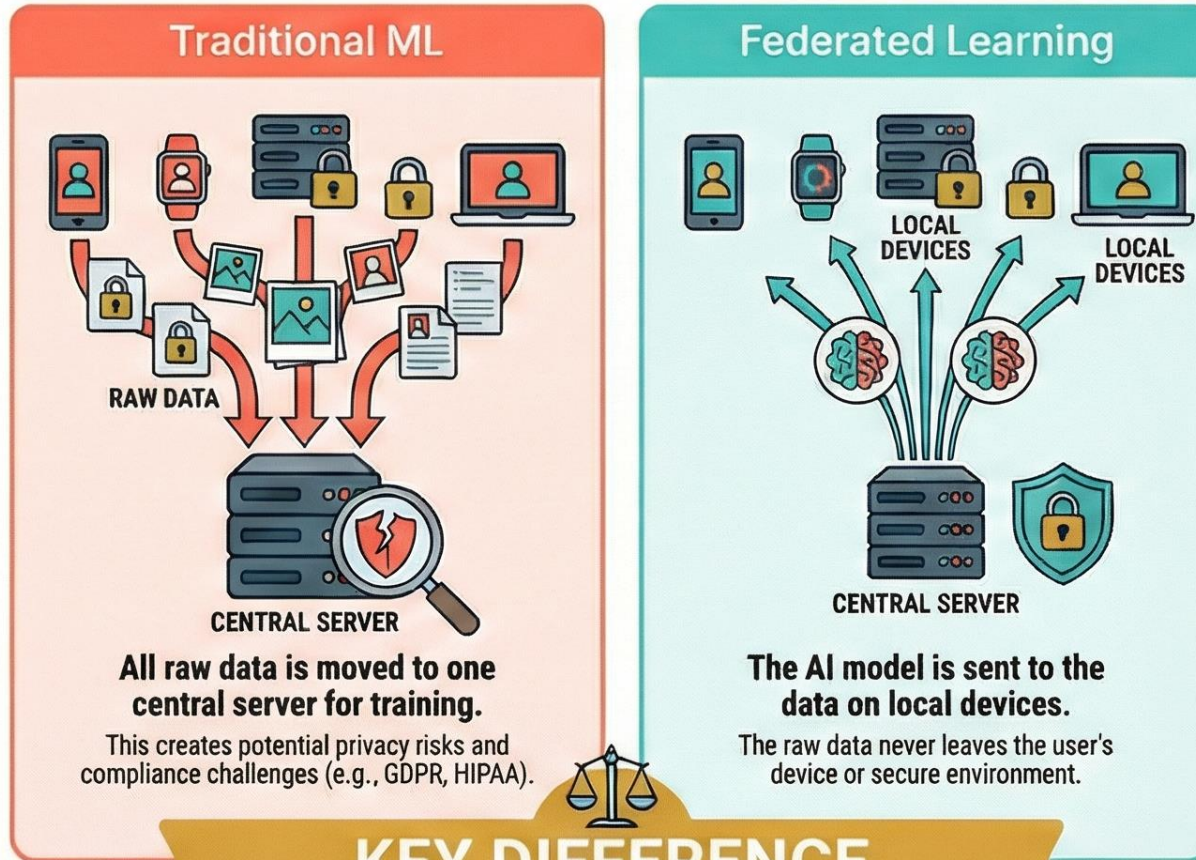
- Data remains local at devices.
- Models are trained collaboratively.
- Only model updates or parameters are exchanged.
- A coordinator aggregates updates to form a global model.

Centralised learning vs Federated Learning

- Centralised learning assumes full data access.
- Federated learning assumes partial and private data.
- Communication replaces data transfer.
- Optimisation becomes distributed and approximate.

Federated Learning: Smarter AI Without Sharing Your Data

Traditional vs. Federated Machine Learning



RAW DATA

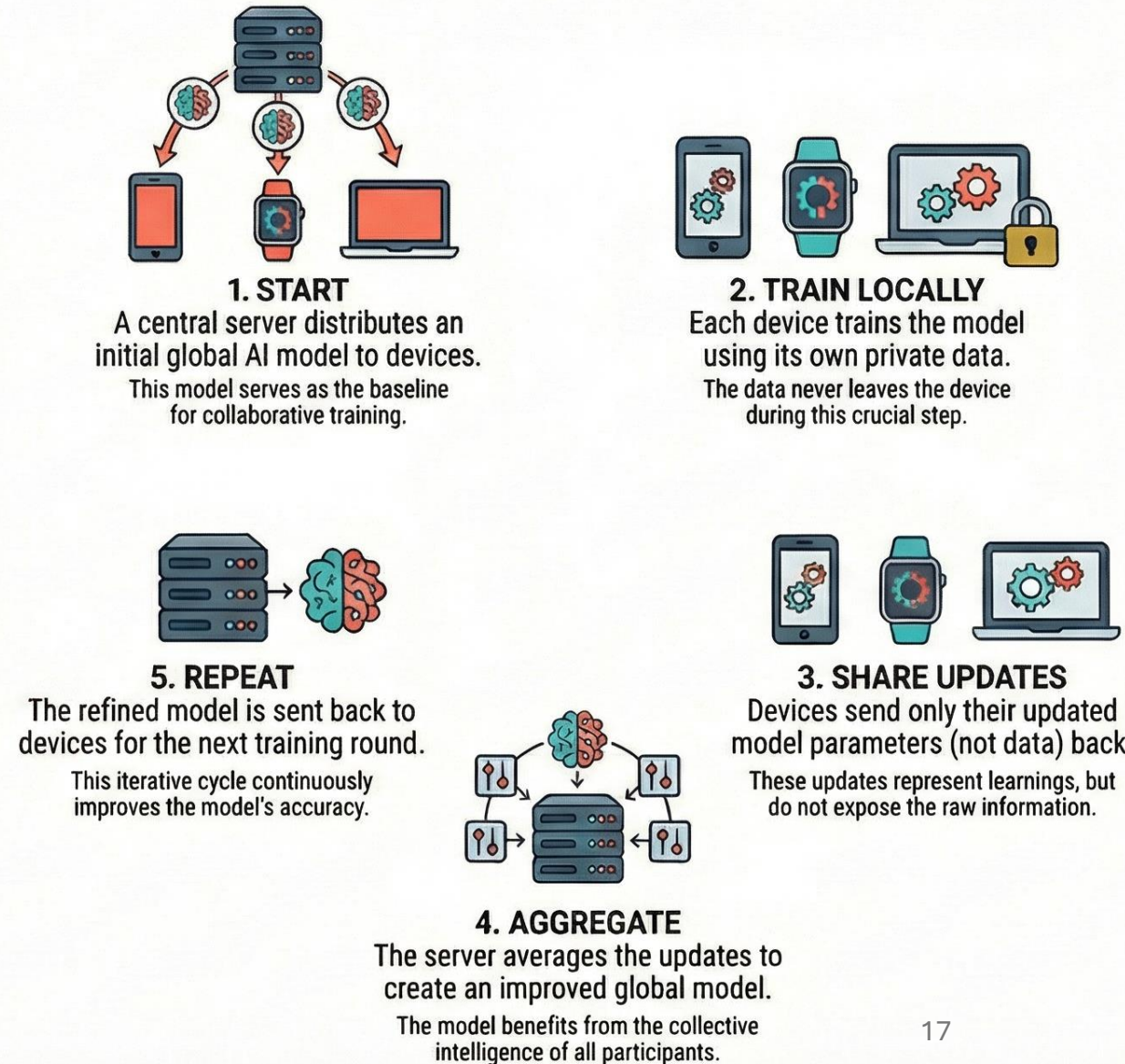
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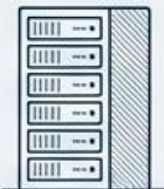
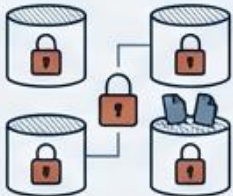
ANONYMOUS MODEL UPDATES

The key difference is what gets shared: raw data vs. anonymous model updates. This allows for collaborative model improvement without sharing sensitive information.

How The Federated Learning Cycle Works



Comparative Analysis: Centralized vs. Decentralized

Feature	Centralized ML (Traditional)	Federated Learning
Data Location	Consolidated on central server. 	Remains local on client devices. 
Privacy Profile	High risk; raw data accessible to aggregator. 	High privacy; only model updates shared. 
Bandwidth Usage	High; transfer of entire raw datasets. 	Low; transmits only small parameter updates. 
Data Access	Limited by silos and confidentiality. 	Access to private, diverse real-world data. 

FL Mathematical Formulation

Global optimisation objective

Minimise a weighted sum of local loss functions.

Each client contributes proportionally to its data size.

Global objective is never directly observed.

$$\min_w F(w) = \sum_{k=1}^K p_k F_k(w)$$

Local optimisation at clients

Each client performs local stochastic gradient descent.

Multiple local steps approximate the global gradient.

Approximation error depends on data heterogeneity.

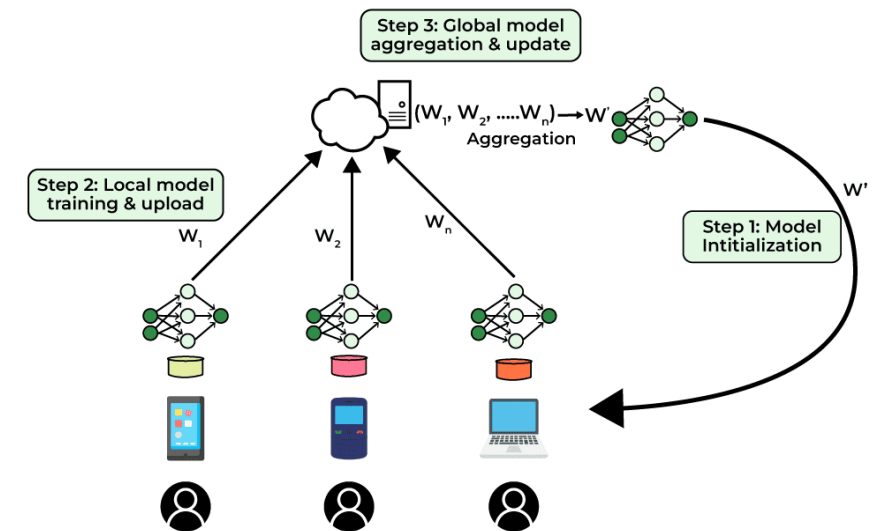
Role of the server

Initialise and broadcast the global model.

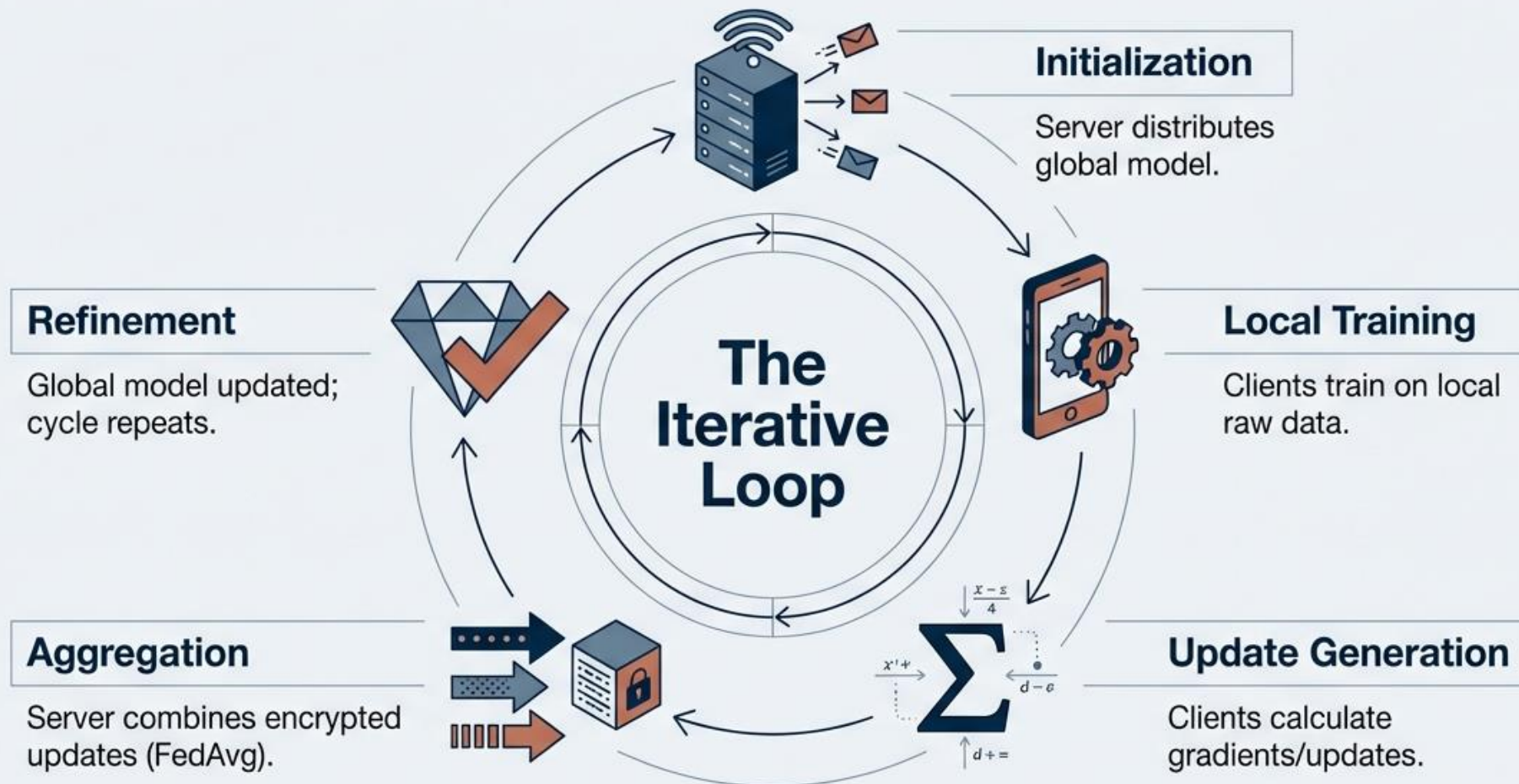
Coordinate training rounds.

Aggregate local updates.

Redistribute the updated model.



FL Iterative Cycle



Part 4 – Federated Learning under Real IoT Constraints

FL under Real IoT Constraints

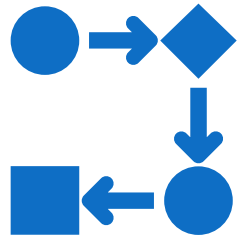
Statistical heterogeneity (Non-IID data)

- Devices observe different data distributions.
- Local objectives differ significantly.
- Causes divergence between client updates.
- Slows or destabilises convergence.

Device and system heterogeneity

- Devices have different compute capabilities.
- Energy budgets vary across clients.
- Network connectivity is intermittent.
- Participation is often partial.

Practical Problems and Constraints



Client selection strategies

Random selection is simple but inefficient.

Resource-aware selection improves efficiency.

Utility-aware selection improves model quality.

Trade-off between fairness and convergence speed.



Communication constraints

Model updates are large.

Bandwidth is limited at the edge.

Frequent communication increases energy consumption.

Advanced Federated Learning methods



FedProx

- Adds a proximal term to local objectives.
- Limits deviation from the global model.
- Improves stability under heterogeneity.
- Slightly increases computation cost.

SCAFFOLD

- Introduces control variates.
- Reduces variance in local updates.
- Corrects client drift explicitly.
- Improves convergence for non-IID data.

FL Algorithms

FedAvg (Baseline)

Standard averaging after E local epochs.

$$\mathbf{w}^{(t+1)} \leftarrow \sum \frac{n_k}{n} \cdot \mathbf{w}_k^{(t+1)}$$

⚠ Diverges in heterogeneous (Non-IID) settings due to client drift.

FedProx (Stability)

Adds a proximal term to penalize deviation.

$$\min F_k(\mathbf{w}) + \frac{\mu}{2} \|\mathbf{w} - \mathbf{w}^t\|^2$$

🟢 Superior stability; mitigates straggler effects.

SCAFFOLD (Drift Correction)

Uses control variates to correct drift direction.

$$\mathbf{w}_k^{(t+1)} \leftarrow \mathbf{w}_k^t - \eta(\mathbf{g}_k - \mathbf{c}_k + \mathbf{c})$$

🟢 Outperforms in highly Non-IID scenarios (e.g., 89% vs 84% accuracy).

Statistical Heterogeneity (Non-IID): When local data distributions differ significantly from the global population (e.g., a failing pump vs. a new pump).

Part 5 - Privacy, Security, and Trust in Collaborative Edge Intelligence

Privacy, Security, and Trust in Collaborative Edge Intelligence

Why trust is a first-class system concern

- AIoT systems interact directly with the physical world.
- Decisions can have safety and economic consequences.
- Loss of trust undermines system adoption.
- Privacy constraints shape architecture choices.

Expanded attack surface in AIoT

- Large number of distributed edge devices.
- Limited physical protection of terminals.
- Heterogeneous software stacks.
- Increased exposure compared to centralised systems.

Privacy risks in Federated Learning



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Gradient and update leakage

Model updates may encode sensitive information.
Adversaries can infer training data properties.
Risk increases with high-dimensional models.



Inference and reconstruction attacks

Attackers reconstruct samples from gradients.
Membership inference identifies participation.
Privacy risk persists without raw data sharing.



Threat model assumptions

Honest-but-curious server.
Malicious or compromised clients.
Limited computational capability at edge.

Privacy-preserving mechanisms

Differential privacy in FL

- Noise added to gradients or updates.
- Formal privacy guarantees.
- Trade-off between privacy and accuracy.
- Careful tuning required.

Secure aggregation

- Server only sees aggregated updates.
- Individual updates are hidden.
- Protection against honest-but-curious servers.

Privacy-preserving mechanisms



Cryptographic techniques

- Homomorphic encryption for computation on encrypted data.
- Secure multi-party computation concepts.
- Often too heavy for edge devices.

Practical privacy-by-design principles

- Minimise shared information.
- Prefer local inference over training when possible.
- Combine system-level and algorithm-level defences.

Data poisoning attacks

- Malicious clients inject corrupted data.
- Model accuracy degrades globally.
- Hard to detect in distributed settings.

Model poisoning attacks

- Attackers manipulate model updates directly.
- Can introduce targeted backdoors.
- High impact with small number of attackers.

Robust aggregation strategies

- Median and trimmed mean aggregation.
- Anomaly detection on updates.
- Trade-off between robustness and efficiency.

Limitations of defences

- Increased computation and communication cost.
- Reduced convergence speed.
- No single defence is sufficient.

Part 6 – Case Studies, Tools, and Project Alignment

Applications: Mobile Experiences & Personalisation



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Predictive Text:

Next-word suggestions (e.g., Gboard) learn from typing styles locally.



Voice Recognition:

"Hey Google" detection improves on-device.



Privacy-First Apps:

Features adapt to user behavior while keeping PII on the device.

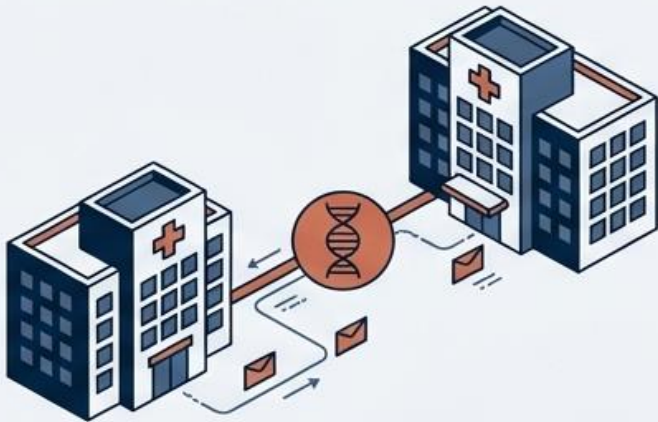
Applications: Healthcare, Finance, Industrial IoT



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Healthcare

Hospitals collaborate on cancer detection without sharing patient records (HIPAA compliant).



Finance

Banks join forces on anti-money laundering (AML) without exposing customer PII.



IoT & Industrial

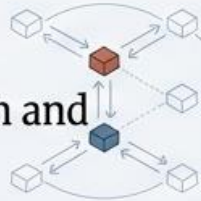
Factories run predictive maintenance on edge equipment to protect proprietary sensor data.



Frameworks and Tools



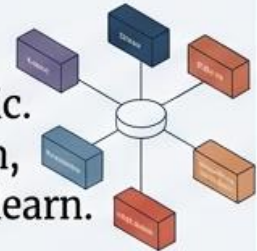
Google's open-source framework. Simulation and research focus.



OpenMined Python library. Focus on privacy & differential privacy.



Framework-agnostic. Works with PyTorch, TensorFlow, scikit-learn.



Specialized for medical imaging & genomics.



WeBank enterprise platform. Banking-grade security.



Medical focus. Emphasis on data ownership and traceability.



The Open Source Toolkit for Edge Intelligence.

Course Project Cheat Sheet.

Category	Tool	Key Feature
Framework	EdgeX Foundry	Vendor-neutral device integration.
AI / Inference	TensorFlow Lite	Optimized for edge/microcontrollers.
Orchestration	K3s & KubeEdge	Lightweight K8s with offline autonomy.
Messaging	Eclipse Mosquitto	MQTT Standard.
Privacy	PySyft	Encrypted computation.
Development	Flower & Node-RED	FL Framework & Visual Workflow.

Looking ahead



Suggested project list of Week 3



Finalise group formation and presentation slot (in Week 4)



Week 4-5: Connectivity and Protocols in IoT Systems.

References and Helpful Links

- [1] Wu J, Liu J, Liu Y, Wang L, Wang Z, Chen W, Tian Z, Yu R, Leung V. A Survey on Cloud-Edge-Terminal Collaborative Intelligence in AIoT Networks. arXiv preprint arXiv:2508.18803. 2025 Aug 26.
- [2] McMahan B, Moore E, Ramage D, Hampson S, y Arcas BA. Communication-efficient learning of deep networks from decentralized data. In Artificial intelligence and statistics 2017 Apr 10, pp. 1273-1282, PMLR.
- [3] B. Ali, M. Golec, S. S. Gill, F. Cuadrado and S. Uhlig, "EdgeAIBus: AI-Driven Joint Container Management and Model Selection Framework for Heterogeneous Edge Computing," in *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 11, pp. 2412-2424, Nov. 2025, doi: 10.1109/TPDS.2025.3602521
- [4] Flower framework tutorial: <https://flower.ai/docs/framework/tutorial-series-what-is-federated-learning.html>
- [5] PySyft: <https://github.com/OpenMined/PySyft>
- [6] TensorFlow Federated: <https://www.tensorflow.org/federated>

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